Supplement 11-Integrating Factor

Show that the given function is an Integrating Factor and then solve the differential equation.

1. $2\cos\pi y\, dx = \pi \sin \pi y\, dy$

   $f = e^{2x}$

2. $(\frac{3}{2}y^2 + 3y + 4x^2)dx + x(y+1)dy = 0$

   $f = x^2$

3. $(4xy + 6y^2)dx + (2x^2 + 6xy)dy = 0$

   $f = x^2 y$

Solve the initial value problem using the given Integrating Factor.

1. $(x - 2y^2)dx - 2xydy = 0;\ y(-2) = 4$

   $f = x$